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Bihar Engineering University, Patna

B.Tech 8th Semester Exam., 2025

Code: 105816 | ***Time: 03 Hours***

Semester: VIII | ***Subject: EMBEDDED SYSTEMS*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only):- ***[2 x 7 = 14]***

(a) Which of the following processors can perform exponential, logarithmic and trigonometric functions?
(i) 8086 (iii) 8080
(ii) 8087 (iv) 8088

(b) How many stack register does an 8087 have?
(i) 4 (iii) 16
(ii) 8 (iv) 32

(c) How are negative numbers stored in a coprocessor?
(i) 1's complement (iii) decimal
(ii) 2's complement (iv) gray

(d) How is memory accessed in RISC architecture?
(i) load and store instruction (iii) memory instruction
(ii) opcode instruction (iv) bus instruction

(e) Princeton architecture is also known as :
(i) Harvard architecture (iii) RISC
(ii) Von Neumann architecture (iv) CISC

(f) What is the use of address bus?
(i) to provide data to and from the chip (iii) to select a location within the memory chip
(ii) to select a specified chip (iv) to select a read/write cycle

(g) Which of the following can affect the long distance communication?
(i) clock (iii) inductor
(ii) resistor (iv) capacitor

(h) What is the use of address bus?
(i) to provide data to and from the chip (iii) to select a location within the memory chip
(ii) to select a specified chip (iv) to select a read/write cycle

(i) Which of the following can transfer multiple bits of data simultaneously?
(i) serial port (iii) concurrent unit
(ii) sequential port (iv) parallel port

(j) Which is the most basic non-volatile memory?
(i) Flash memory (iii) EPROM
(ii) PROM (iv) ROM

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Q.2 ***[7+7=14]***

(a) With a neat sketch explain ARM programming model. What do you mean by pipelining? Briefly discuss about five stage pipeline in ARM.

(b) Discuss the co-processor Register transfer instructions. Why the instruction cannot be used for Register transfer of CP15 co-processor? Justify.

Q.3 ***[7+7=14]***

(a) With a neat diagram discuss set associate cache and fully associative cache.

(b) What are the different types of memories used in embedded system design? Explain each with examples.

Q.4 ***[7+7=14]***

(a) Difference between instruction set and thumb instruction set with suitable example.

(b) What are the different types of timing analyses techniques? Discuss them in detail.

Q.5 ***[7+7=14]***

(a) Differentiate between memory protection unit(MPU) and memory management unit(MMU).

(b) What is compilation? What are the different types of compilation? Discuss them in detail.

Q.6 ***[7+7=14]***

(a) Discuss the use of controllers in embedded systems.

(b) Formulate necessary code using ARM assembly language program for creating a delay.

Q.7 ***[7+7=14]***

(a) Explain various data processing instruction with their syntax .

(b) Draw and compare von-Neumann and Harvard architecture.

Q.8 ***[7+7=14]***

(a) How can you improve performance by scheduling of load instructions? Explain it in detail.

(b) Discuss about the ARM C compiler and its optimization techniques.

Q.9 Write short notes on any two of the following:- ***[7 x 2 = 14]***

(a) CPU performance & Power Consumption trade-off

(b) SHARC Architecture

(c) Component Interfacing

(d) Linking

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